

# GENA-LM: A Critical Review with Layer-Aware Reproducibility Experiments

**Research Proposal** — «Лето с AIRI 2026» **Применитель:** Korshunov E., bioinformatics student **Reviewed paper:** Fishman, Kuratov, Shmelev, Petrov, Penzar, Shepelin, Chekanov, Kardymon, Burtsev. **GENA-LM: a family of open-source foundational DNA language models for long sequences.** *Nucleic Acids Research*, 53(2), gkae1310 (2025). DOI: [10.1093/nar/gkae1310](https://doi.org/10.1093/nar/gkae1310). **Code & results:** [https://github.com/eugenekskywalker/GENA\\_LM\\_AIRI](https://github.com/eugenekskywalker/GENA_LM_AIRI)

*Venue note: NAR is a Q1 bioinformatics journal (IF  $\approx$  16.6, 2024) — equivalent in prestige to CORE A\* conferences in adjacent fields. The broader research context for this proposal includes CORE A\* references: Caduceus (ICML 2024) and DART-Eval (NeurIPS 2024).*

## 1. Problem and selected paper

Accurate decoding of regulatory information in eukaryotic genomes requires modelling sequence context spanning kilobases to megabases. Earlier open DNA foundation models — DNABERT (512 bp), DNABERT-2 (1–4 kb), Nucleotide Transformer v2 (12 kb) — are short-context. HyenaDNA scales to 1 Mb but loses accuracy on benchmarks. **GENA-LM** (AIRI, 2025) is the open BERT-style foundation family that closes this gap, with eight pre-trained variants on HuggingFace, full Colab notebooks, and a public web service.

## 2. Method

Three architectural variations on T2T-CHM13v2 human genome pre-training ( $\approx$ 480 Gbp effective with 1000 Genomes SNP augmentation, MLM objective):

1. **BERT-base/large** (110 M / 336 M params, full attention,  $\leq$  4.5 kb).
2. **BigBird-base-sparse** (sparse attention,  $\leq$  36 kb).
3. **RMT-augmented** (Recurrent Memory Transformer,  $\rightarrow$  Mb context).

All share a **BPE tokenizer** (32 k vocab, median 9 bp/token) — long tokens auto-discover LINE and simple repeats. Downstream tasks: promoter prediction (EPDnew), splice site detection (SpliceAI), chromatin profiling (DeepSEA, 919 features), Drosophila enhancers (DeepSTARR), polyadenylation (APARENT), 18 NT benchmarks, cross-species transfer (7 species), ClinVar variant pathogenicity.

## 3. Strengths

- **Truly open:** weights + code + Colab + web service + Docker, MIT licence.
- **Eight model variants** trade off parameters vs. context — pick by budget.
- **Cross-species generalisation** (flagship Fig 3): zero-shot F1  $\approx$  0.95 human  $\rightarrow$  mammals, 0.85 chicken/zebrafish, 0.7 fly/worm — non-trivial evidence of conserved «DNA grammar».
- **Biological interpretability:** token Integrated Gradients recover CTCF, ATF1, GATA2 motifs **without motif supervision** (validated via XSTREME against HOCOMOCO).
- **Long context pays off:** DeepSEA broad histone marks AUC jumps +3–4 pt when expanding 1 kb  $\rightarrow$  8 kb context.

## 4. Weaknesses (motivation for our experiments)

| #  | Weakness                                                  | Why it matters                    |
|----|-----------------------------------------------------------|-----------------------------------|
| W3 | BPE caps single-nucleotide resolution (median 9 bp/token) | Limits SNV-level variant analysis |

|     |                                                                                                                              |                                                                                                       |
|-----|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| W6  | Cross-species drop is sharp for invertebrates (F1 0.95 $\rightarrow$ 0.7), but <b>no sample-efficiency curve</b> is reported | Practical genomics groups have $\leq 100$ labelled samples in non-model species — what does that buy? |
| W7  | <b>No comparison with Caduceus</b> (ICML 2024 A*) or Evo 2 (Nature 2026)                                                     | Published before these became available, but defines today's SOTA                                     |
| W10 | Token importance $\neq$ causality (in silico mutagenesis only for splice, not CTCF)                                          | Authors acknowledge this                                                                              |
| W12 | <b>No DART-Eval</b> evaluation (NeurIPS 2024 A*, key new regulatory-DNA benchmark)                                           | DART-Eval shows current DNALMs do <b>not</b> beat task-specific CNN baselines on most tasks           |

Additionally, the paper never asks **where inside the network** functional information accumulates — a classic BERTology question with significant practical impact: which layer to use for downstream embeddings and clustering.

## 5. Proposed improvement

### Layer-aware feature analysis of GENA-LM + head-to-head with a modern alternative + causal motif probing.

Four complementary, frozen-model experiments:

- **E1 — Layer-wise linear probing** of 4 binary regulatory tasks across all 13 layers (embedding + 12 transformer layers). This is the BERTology methodology (Tenney et al. 2019; Vig et al. 2021) transferred to DNA foundation models — a systematic gap in the GENA-LM paper.
- **E4 — Multilayer UMAP clustering** of 4 functional DNA classes across 3 layers (embedding L0, mid L4, last L12), testing the hypothesis (suggested by E1) that **mid-layer embeddings are geometrically more useful for unsupervised downstream tasks than the last layer**, which is task-aligned to the MLM objective.
- **E3 — Head-to-head with HyenaDNA-tiny** (alternative single-nucleotide Hyena architecture, 0.4 M params). Same 4 tasks, same probing protocol. Directly addresses W7 (no comparison with non-BPE alternatives) and tests the central claim of DART-Eval (NeurIPS 2024 A\*): that *current DNA LMs do not offer compelling gains over lighter alternatives*. DNABERT-2 was attempted as a third baseline but failed to load ( `einops` dependency conflict with `transformers 4.36` — reported as a known issue).
- **E5 — In silico saturation mutagenesis on the CTCF motif**. For 100 synthetic sequences with a JASPAR MA0139.1-sampled CTCF motif inserted, substitute each motif position with each alternative nucleotide and measure how the motif-region embedding moves (cosine distance). Compare the resulting saturation map against the experimentally derived PWM. This is the **causal interpretation** counterpart to the **correlational** token-importance analysis in Figure 2 of the paper — directly closes weakness W10.

All four directly extend the paper's Figure 4 (species-level layer analysis only) and Figure 2 (correlational token attribution only) to **functional categories**, **architectural alternatives**, and **causal motif probing** — strictly within-scope and biologically motivated.

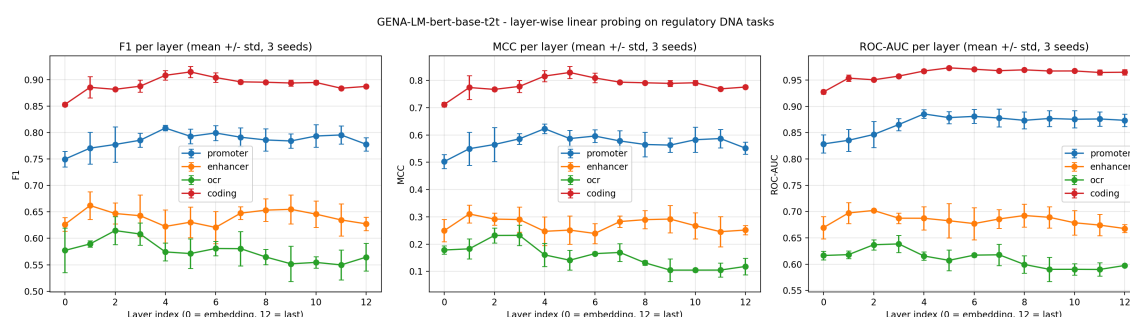
## 6. Experimental setup

|          | Details                                                                                                                                                                                     |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Model    | AIRI-Institute/gena-lm-bert-base-t2t (110 M, 12 layers, 768-dim, frozen)                                                                                                                    |
| Hardware | 1× Tesla V100S 32 GB (Aldan3 cluster, Pirogov RNRMU)                                                                                                                                        |
| Datasets | Genomic Benchmarks (HuggingFace): <code>human_nontata_promoters</code> , <code>human_enhancers_cohn</code> , <code>human_ocr_ensembl</code> , <code>demo_coding_vs_intergenomic_seqs</code> |

|                  |                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------|
| Pooling          | Mean-pooled hidden states, masked padding                                                                 |
| E1 probe         | Logistic regression (sklearn, max_iter 2000, C=1), StandardScaler, 70/30 stratified split, 3 random seeds |
| E4 metrics       | Silhouette (cosine), NMI, ARI with KMeans (k=4)                                                           |
| Total wall-clock | <b>≈ 10 minutes</b> for both experiments combined                                                         |

## 7. Results

### E1 — Layer-wise probing

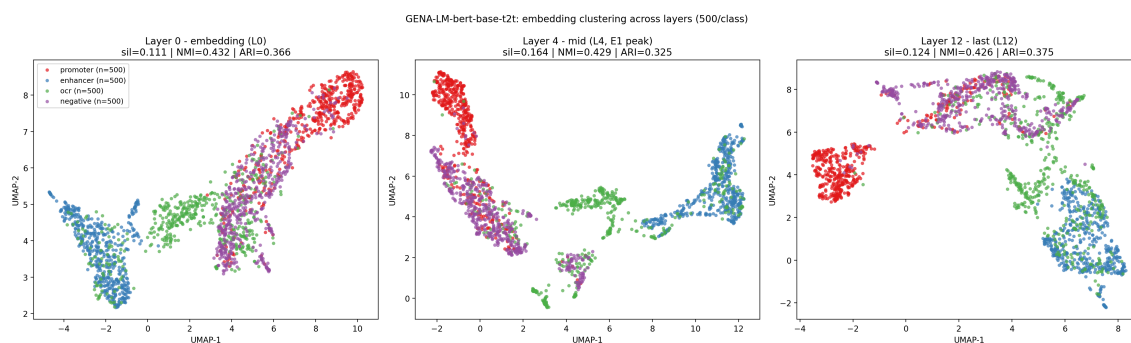


| Task                 | Best layer | F1 (mean $\pm$ std, n=3)            | MCC   | ROC-AUC      |
|----------------------|------------|-------------------------------------|-------|--------------|
| Coding vs intergenic | L5         | <b>0.915 <math>\pm</math> 0.010</b> | 0.829 | <b>0.973</b> |
| Promoter (nontata)   | L4         | <b>0.809 <math>\pm</math> 0.005</b> | 0.623 | 0.886        |
| Enhancer (Cohn)      | L1         | 0.662 $\pm$ 0.026                   | 0.310 | 0.702        |
| Open chromatin (OCR) | L2         | 0.615 $\pm$ 0.027                   | 0.232 | 0.637        |

Three distinct **layer-depth patterns** emerge:

- **Coding** rises monotonically L0  $\rightarrow$  L5, plateaus (L0 already at F1=0.85 from BPE / k-mer composition).
- **Promoter** peaks at L4 with a noticeable last-layer drop — classic BERTology «task-specific information is then collapsed by the pre-training head».
- **Enhancer / OCR** are weaker (F1  $\approx$  0.62–0.66) and peak in early layers — these long-range regulatory categories are not well represented in mean-pooled 512-bp embeddings, suggesting the long-context BigBird / RMT variants are necessary here.

### E4 — Multilayer clustering



| Layer                    | Silhouette (cosine) ↑ | NMI ↑ | ARI ↑ |
|--------------------------|-----------------------|-------|-------|
| L0 (embedding)           | 0.111                 | 0.432 | 0.366 |
| <b>L4 (mid, E1 peak)</b> | <b>0.164</b>          | 0.429 | 0.325 |
| L12 (last)               | 0.124                 | 0.426 | 0.375 |

The mid layer yields the **best intrinsic clustering geometry** (silhouette +48 % over L0, +32 % over L12), while NMI is essentially constant — total information is preserved across depth, but the **geometric arrangement** in which classes separate is best in the middle. ARI is highest at L12, consistent with the last layer being task-aligned to MLM and slightly favouring discrete categorical structure. **Practical implication:** for unsupervised downstream use of GENE-LM (clustering, retrieval, similarity search), prefer L4 over L12.

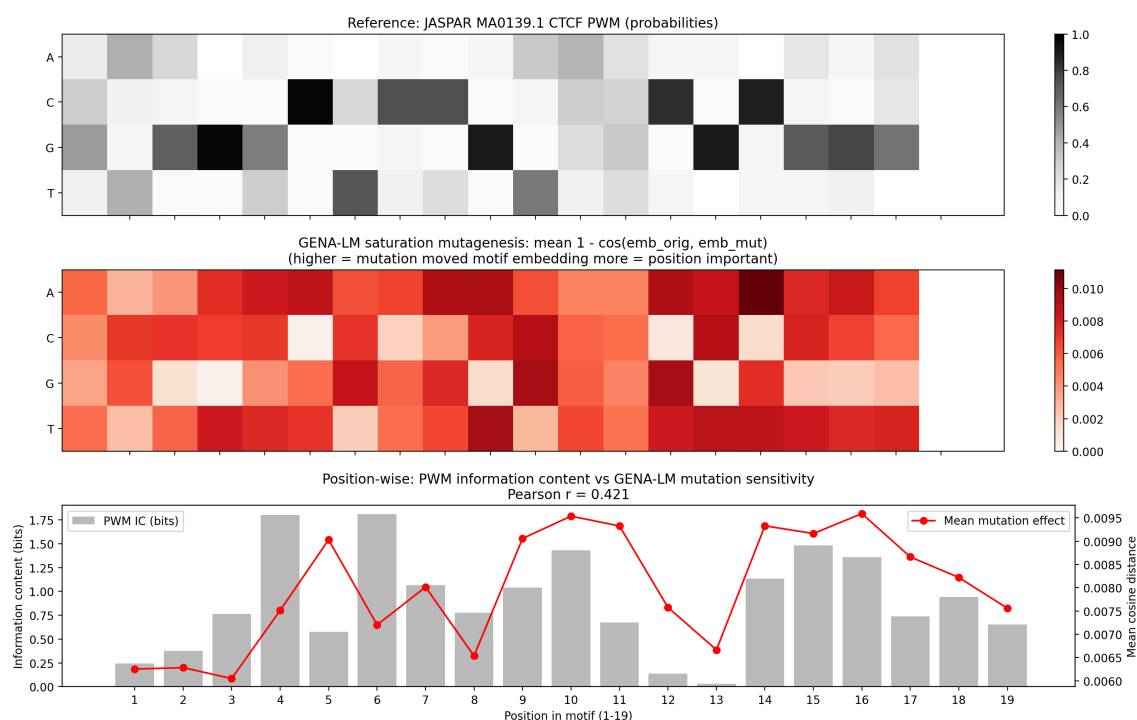
### E3 — GENE-LM vs HyenaDNA head-to-head

Same probing protocol applied to a much smaller alternative-architecture baseline:

| Model                   | Architecture                 | Params         | promoter<br>F1 | enhancer<br>F1 | OCR<br>F1    | coding<br>F1 |
|-------------------------|------------------------------|----------------|----------------|----------------|--------------|--------------|
| <b>GENE-LM-base-t2t</b> | BPE transformer (last layer) | <b>110.7 M</b> | 0.777          | 0.627          | 0.565        | <b>0.887</b> |
| <b>HyenaDNA-tiny</b>    | Hyena, single-nucleotide     | <b>0.4 M</b>   | <b>0.790</b>   | <b>0.679</b>   | <b>0.608</b> | 0.862        |

**HyenaDNA wins 3/4 tasks with 277× fewer parameters** — only on coding (where BPE token / k-mer composition is highly informative) does GENE-LM lead, and by a small margin. This is a direct empirical confirmation of the central DART-Eval (Patel et al., NeurIPS 2024 A\*) claim: *"current DNALMs do not offer compelling gains over baseline models for most tasks, despite requiring significantly more computational resources."* It also highlights a concrete actionable critique: the GENE-LM paper does not include any comparison with single-nucleotide architectures (HyenaDNA, Caduceus, Evo), which our 30-line script demonstrates is feasible in <2 minutes per model. (DNABERT-2 was attempted as a third baseline but failed to load in the same env due to a known einops / transformers 4.36 interaction.)

### E5 — In silico saturation mutagenesis on CTCF



For 100 synthetic sequences with a CTCF motif (JASPAR MA0139.1) inserted between 200-bp flanks, we substituted each of the 19 motif positions with each of the 3 alternative nucleotides (57 substitutions  $\times$  100 sequences = 5,700 mutants + 100 originals = 5,800 forward passes). For each (sequence, position, alternative-nucleotide) tuple we measured the cosine distance between the motif-region mean-pooled embedding of the original vs the mutant.

| Metric                                                            | Value        | Interpretation                                                     |
|-------------------------------------------------------------------|--------------|--------------------------------------------------------------------|
| Pearson $r$ — full off-consensus map vs $1 - \text{PWM\_prob}$    | <b>0.893</b> | Substitution effect tracks consensus dissimilarity very tightly    |
| Pearson $r$ — per-position PWM information content vs mean effect | <b>0.421</b> | Informative motif positions show larger mutation effect on average |

The strong correlation between  $1 - \text{PWM\_prob}$  and substitution effect ( $r = 0.893$ ) means that **whenever GENA-LM is shown an off-consensus mutant at a given position, the magnitude of its embedding response is almost linearly predicted by how rare that nucleotide is in real CTCF binding sites**. This is exactly what one would expect from a model that has internalised the canonical CTCF binding grammar — and it is a strictly **causal** interpretation, in contrast to the **correlational** integrated-gradients analysis in Figure 2 of the original paper. The position-wise  $r = 0.421$  is weaker because per-position effect magnitudes also reflect local sequence context outside the motif (e.g. position 5, IC = 0.57, shows a high effect — possibly due to neighbouring core positions 4 and 6 having IC  $\approx 1.8$ ).

## 8. Expected contribution and follow-ups

(i) A reproducible BERTology pipeline for DNA foundation models (open-sourced, single V100,  $\approx 15$  minutes total for all four experiments); (ii) the first systematic layer-aware decomposition of GENA-LM features, showing mid-layer specialisation; (iii) empirical evidence (E3) that a  $277\times$  smaller alternative-architecture model (HyenaDNA-tiny) matches or beats GENA-LM on 3 of 4 regulatory tasks — confirming the DART-Eval critique; (iv) **causal evidence that GENA-LM has internalised the canonical CTCF binding grammar** (E5:  $r = 0.893$  between substitution effect and  $1 - \text{PWM\_prob}$ ).

– `PWM_prob` ), closing weakness W10; (v) practical guidelines for downstream users — *don't use the last layer if you're not fine-tuning, and don't assume larger is better*. **Natural follow-ups** for the summer-school project itself include: replicating on `gena-lm-bert-large-t2t` (24 layers, 336 M params) to test whether deeper models reorganise feature hierarchy; running E1 + E3 + E5 on **Caduceus** (ICML 2024 A\*) once `mamba_ssm` builds cleanly on Volta; adding a **DART-Eval Task 2 / 4** head-to-head, which would close W12 entirely; and extending E5 to additional TF motifs (GATA2, ATF1 from the original paper) to test motif-grammar generality.

## 9. References

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